

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: pf	Course Code: CMPE-112L
Assignment Type: lab	Dated: 25-05-2026
Semester: 1	Session: Spring 2026
Lab/Project/Assignment #: 13	CLOs to be covered: CLO 2
Lab Title: Map, filter and reduce	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- To understand the use of Python `map()` and `filter()` functions.
- To learn how to use lambda functions with `map()` and `filter()`.
- To understand the concept of exception handling using `try`, `except`, `else`, and `finally`.
- To develop error-handling and functional programming skills in Python.

Equipment Required:

- Laptop/PC
- Python 3.x
- VS Code (or any Python IDE)
- Windows Operating System

Theory:

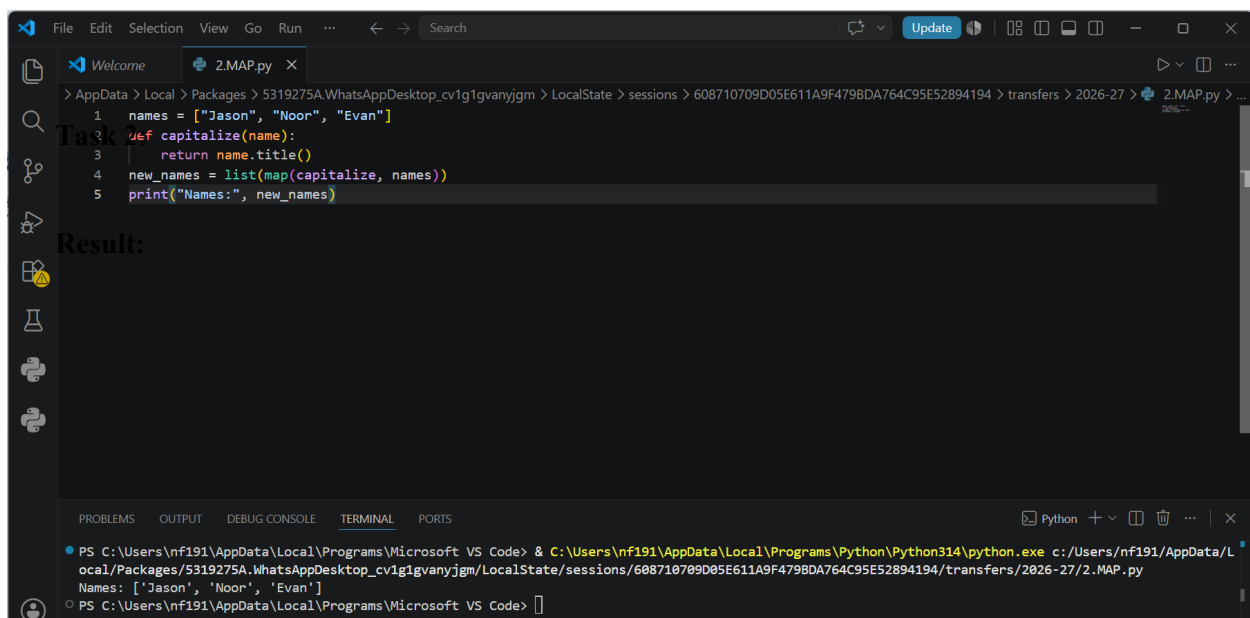
Python provides built-in higher-order functions such as `map()`, `filter()`, and `reduce()` to process data efficiently. The `map()` function applies a specified function to every element of an iterable, `filter()` selects elements that satisfy a given condition, and `reduce()` combines all elements of an iterable into a single value. Python also supports **exception handling** using `try`, `except`, `else`, and `finally` blocks to detect and handle runtime errors, making programs more reliable and preventing unexpected crashes.

Lab Work:

Task 1:

Write a program to use the `map()` function with a user-defined function to capitalize the first letter of each name in a list and display the updated list.

Result:



The screenshot shows a VS Code editor window with a file named `2.MAP.py` open. The code in the editor is as follows:

```
1 names = ["Jason", "Noor", "Evan"]
2 def capitalize(name):
3     return name.title()
4 new_names = list(map(capitalize, names))
5 print("Names:", new_names)
```

Below the editor, the terminal window shows the output of the script:

```
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> C:\Users\nf191\AppData\Local\Programs\Python\Python314\python.exe c:/Users/nf191/AppData/Local/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/608710709D05E611A9F479BDA764C95E52894194/transfers/2026-27/2.MAP.py
Names: ['Jason', 'Noor', 'Evan']
```

Task 2:

Write a program to use the `map()` function with a user-defined function to perform a mathematical calculation on each element of a list and display the resulting values.

Result:

The screenshot shows the Visual Studio Code (VS Code) editor interface. The top menu bar includes File, Edit, Selection, View, Go, Run, and Search. The left sidebar contains icons for Explorer, Search, Source Control, Run and Debug, Extensions, and Testing. The main editor area displays a Python file named `3.map.py` with the following code:

```
1 data = [2, 4, 5]
2 def calc(x):
3     return x**2 + 10
4 output = list(map(calc, data))
5 print("Result:", output)
```

The bottom panel shows the TERMINAL view with the following output:

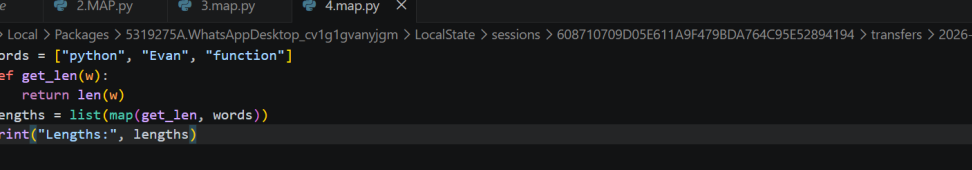
```
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> & C:\Users\nf191\AppData\Local\Programs\Python\Python314\python.exe c:/Users/nf191/AppData/Local/Programs/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/608710709D05E611A9F479BDA764C95E52894194/transfers/2026-27/3.map.py
Result: [14, 26, 35]
```

The status bar at the bottom indicates the current file is `3.map.py` at line 1, column 13, using UTF-8 encoding with CRLF line endings. The Python version is 3.14 (64-bit).

Task 3:

Write a program to use the `map()` function with a user-defined function to find the length of each string in a list and display the resulting list.

Result:



The screenshot shows a VS Code editor with a Python file named `4.map.py` open. The code defines a function `get_len(w)` that returns the length of a word, and then uses `map` and `list` to calculate the lengths of the words "python", "Evan", and "function". The output is printed as "Lengths: [6, 4, 8]". The terminal at the bottom shows the command to run the script, which has been executed successfully.

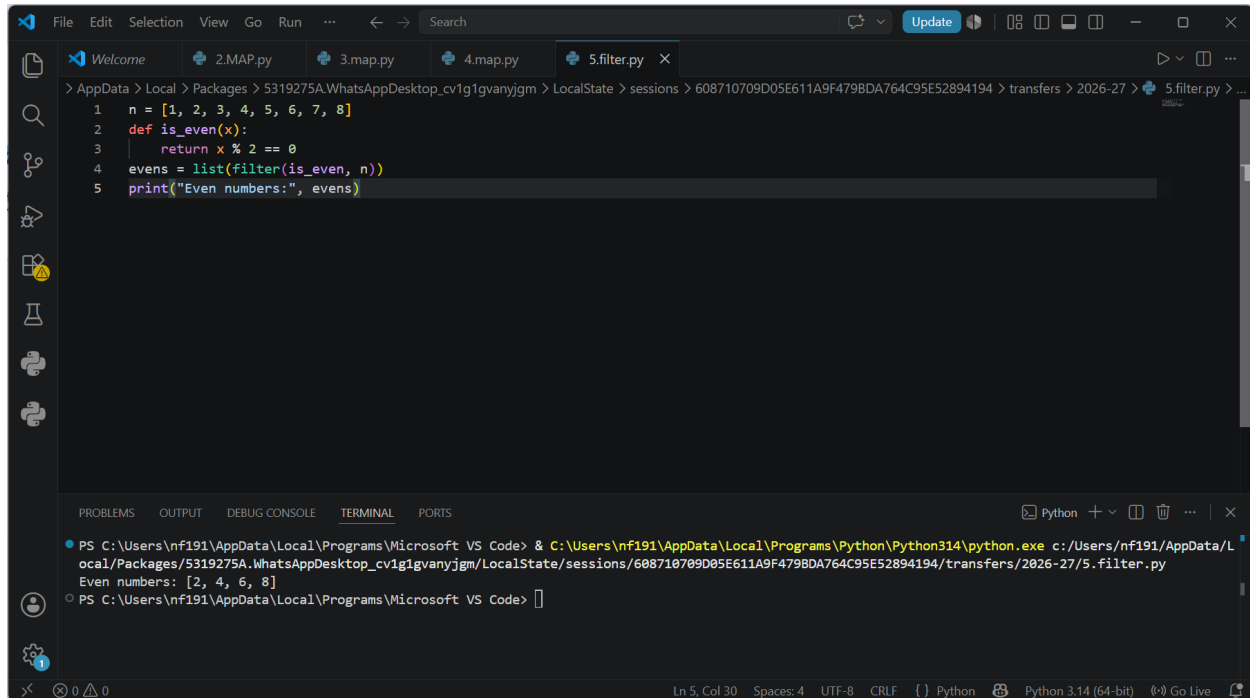
```
1 words = ["python", "Evan", "function"]
2 def get_len(w):
3     return len(w)
4 lengths = list(map(get_len, words))
5 print("Lengths:", lengths)
```

```
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> & C:\Users\nf191\AppData\Local\Programs\Python\Python314\python.exe c:/Users/nf191/AppData/Local/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/608710709D05E611A9F479BDA764C95E52894194/transfers/2026-27/4.map.py
Lengths: [6, 4, 8]
```

Task 4:

Write a program to use the `filter()` function with a user-defined function to extract all even numbers from a list and display the filtered result.

Result:



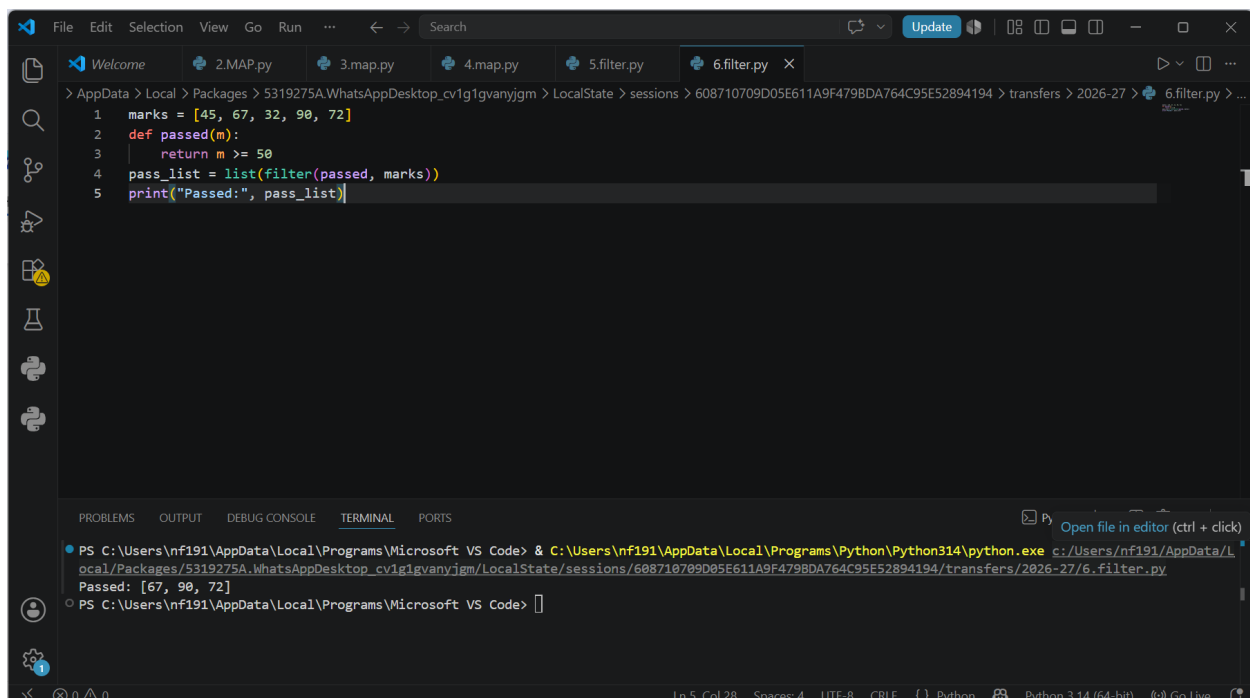
```
File Edit Selection View Go Run ... Search Update
Welcome 2.MAP.py 3.map.py 4.map.py 5.filter.py X
> AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > 608710709D05E611A9F479BDA764C95E52894194 > transfers > 2026-27 > 5.filter.py > ...
1 n = [1, 2, 3, 4, 5, 6, 7, 8]
2 def is_even(x):
3     return x % 2 == 0
4 evens = list(filter(is_even, n))
5 print("Even numbers:", evens)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - Python 3.14 (64-bit) Go Live
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> & C:\Users\nf191\AppData\Local\Programs\Python\Python314\python.exe c:/Users/nf191/AppData/L
ocal/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/608710709D05E611A9F479BDA764C95E52894194/transfers/2026-27/5.filter.py
Even numbers: [2, 4, 6, 8]
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> |
```

Task 5:

Write a program to use the `filter()` function with a user-defined function to filter and display the marks of students who have passed based on a given condition.

Result:



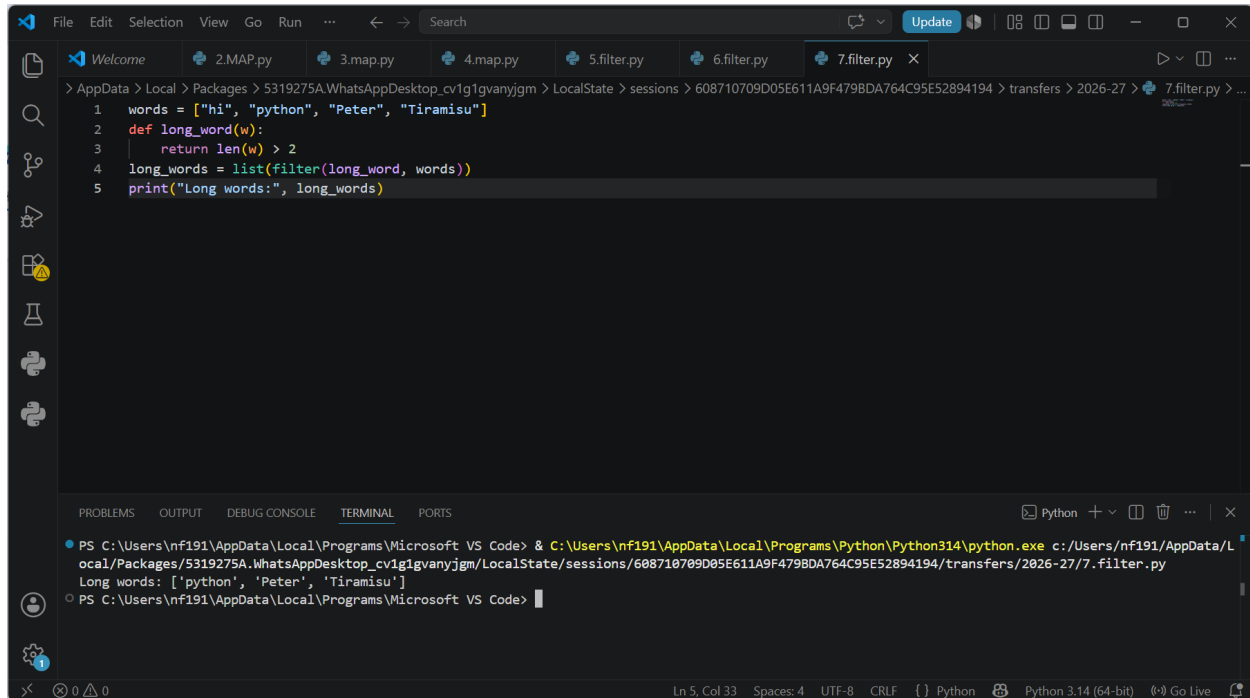
```
File Edit Selection View Go Run ... Search Update
Welcome 2.MAP.py 3.map.py 4.map.py 5.filter.py 6.filter.py X
> AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > 608710709D05E611A9F479BDA764C95E52894194 > transfers > 2026-27 > 6.filter.py > ...
1 marks = [45, 67, 32, 90, 72]
2 def passed(m):
3     return m >= 50
4 pass_list = list(filter(passed, marks))
5 print("Passed:", pass_list)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - Python 3.14 (64-bit) Go Live
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> & C:\Users\nf191\AppData\Local\Programs\Python\Python314\python.exe c:/Users/nf191/AppData/L
ocal/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/608710709D05E611A9F479BDA764C95E52894194/transfers/2026-27/6.filter.py
Passed: [67, 90, 72]
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> |
```

Task 6:

Write a program to use the `filter()` function with a user-defined function to filter and display the words whose length is greater than two characters.

Result:



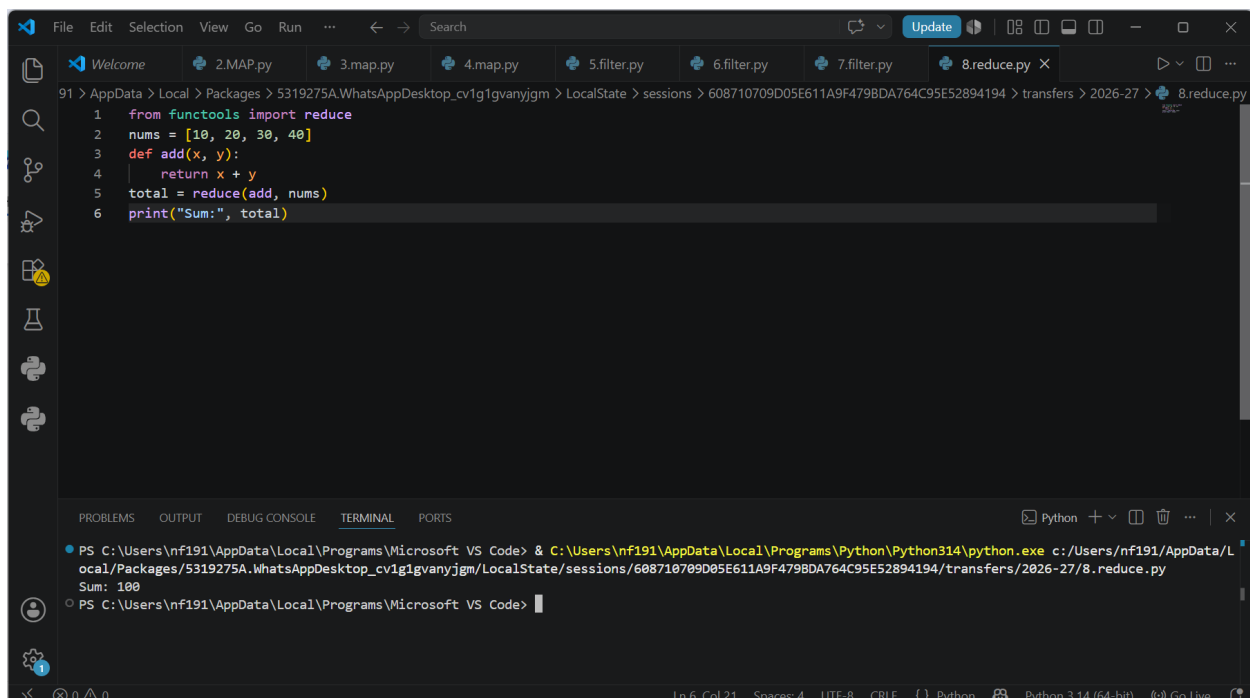
```
1 words = ["hi", "python", "Peter", "Tiramisu"]
2 def long_word(w):
3     return len(w) > 2
4 long_words = list(filter(long_word, words))
5 print("Long words:", long_words)
```

```
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> & C:\Users\nf191\AppData\Local\Programs\Python\Python314\python.exe c:/Users/nf191/AppData/Local/Programs/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/608710709D05E611A9F479BDA764C95E52894194/transfers/2026-27/7.filter.py
Long words: ['python', 'Peter', 'Tiramisu']
```

Task 7:

Write a program to use the `reduce()` function with a user-defined function to calculate the sum of all elements in a list and display the result.

Result:



```
1 from functools import reduce
2 nums = [10, 20, 30, 40]
3 def add(x, y):
4     return x + y
5 total = reduce(add, nums)
6 print("Sum:", total)
```

```
PS C:\Users\nf191\AppData\Local\Programs\Microsoft VS Code> & C:\Users\nf191\AppData\Local\Programs\Python\Python314\python.exe c:/Users/nf191/AppData/Local/Programs/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/608710709D05E611A9F479BDA764C95E52894194/transfers/2026-27/8.reduce.py
Sum: 100
```

Conclusion:

In this lab, we learned how to use the `map()`, `filter()`, and `reduce()` functions with user-defined functions to perform different operations on lists. We also understood the importance of exception handling and how `try`, `except`, `else`, and `finally` blocks help manage runtime errors effectively. This lab enhanced our understanding of functional programming concepts and improved our ability to write efficient and error-free python programs.